

21 Lecture 21: Phase estimation¹

21.1 The circuit²

Consider the eigenvalue equation for a unitary operator U :³

$$U|u\rangle = e^{2\pi i\varphi}|u\rangle \quad (21.1)^4$$

where U and the eigenvector $|u\rangle$ are known, and φ is unknown. The QFT provides⁵ an algorithm to find the phase φ , assuming that black boxes (circuits) are available to prepare the state $|u\rangle$ and to implement controlled- U^{2^j} .

The procedure of phase estimation uses two registers: the **first register** contains t ⁶ qubits, each in the initial state $|0\rangle$. The number t will depend on:

- the accuracy we want to reach in the estimation of the phase⁷
- the probability of success of the algorithm

The **second register** contains initially the eigenstate $|u\rangle$, with the appropriate⁸ number of qubits necessary to encode the eigenstate.

The circuit for the phase estimation is given in Fig. 21.1. The effect of the⁹ sequence of controlled- U gates on the state $|j\rangle|u\rangle$ is

$$|j_1 \cdots j_t\rangle|u\rangle \longrightarrow |j_1 \cdots j_t\rangle U^{2^{t-1}j_1} \cdots U^{2^0j_t}|u\rangle = |j\rangle U^j|u\rangle \quad (21.2)^{10}$$

and recalling the eigenvalue equation (21.1):¹¹

$$|j\rangle|u\rangle \longrightarrow |j\rangle e^{2\pi i\varphi j}|u\rangle \quad (21.3)^{12}$$

The first register exits the H gates in the state:¹³

$$|0\rangle \longrightarrow \frac{1}{2^{\frac{t}{2}}} \sum_{j=0}^{2^t-1} |j\rangle \quad (21.4)^{14}$$

(remember $(|0\rangle+|1\rangle)(|0\rangle+|1\rangle) \cdots (|0\rangle+|1\rangle) = \sum_{j_1=0}^1 \cdots \sum_{j_t=0}^1 |j_1 \cdots j_t\rangle = \sum_{j=0}^{2^t-1} |j\rangle$).¹⁵

Thus the initial state of the two registers $|0\rangle|u\rangle$ is transformed by the circuit as:¹⁶

$$|0\rangle|u\rangle \longrightarrow \frac{1}{2^{\frac{t}{2}}} \sum_{j=0}^{2^t-1} e^{2\pi i\varphi j} |j\rangle|u\rangle = \frac{1}{2^{\frac{t}{2}}} \sum_{j=0}^{2^t-1} e^{2\pi i \frac{(2^t\varphi)j}{2^t}} |j\rangle|u\rangle \quad (21.5)^{17}$$

Suppose now that the phase φ can be expressed *exactly* with t bits:¹⁸

$$\varphi = 0.\varphi_1 \dots \varphi_t \quad (21.6)^{19}$$

(remember that the phase as defined in (21.1) is a real number between 0 and 1).²⁰ Then applying the inverse QFT to the first register gives

$$\frac{1}{2^{\frac{t}{2}}} \sum_{j=0}^{2^t-1} e^{2\pi i \frac{(2^t\varphi)j}{2^t}} |j\rangle \longrightarrow |2^t\varphi\rangle = |\varphi_1 \dots \varphi_t\rangle \quad (21.7)^{21}$$

and measuring the register in the computational basis yields the phase φ .¹

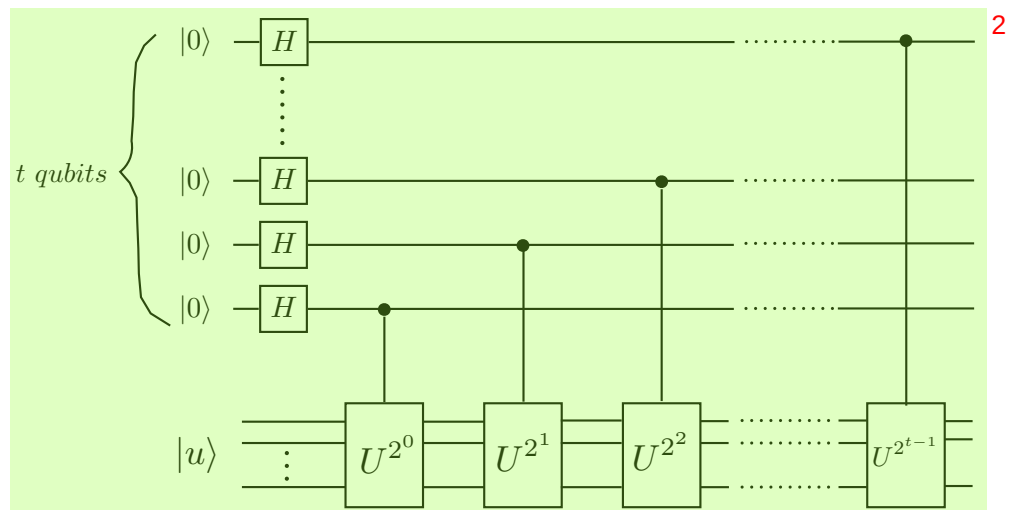


Fig. 21.1 The circuit for phase estimation.³